# Product Requirements Document (PRD)

## SignBridge AI

**Version:** 1.0

**Theme:** Digitally Enabled Inclusive Education: Expanding Access, Equity, and Opportunity for All

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# Executive Summary

SignBridge AI is an AI-powered inclusive education platform designed to remove communication and learning barriers for deaf, hard-of-hearing, and diverse learners.

The platform combines real-time speech transcription, AI-assisted learning, sign language recognition, contextual learning support, and accessibility tools into one intelligent classroom assistant.

Rather than functioning as only a captioning application or only a sign language translator, SignBridge AI creates an inclusive digital learning environment where students can receive live captions, AI-generated explanations, lecture summaries, visual learning cues, and real-time communication support regardless of their hearing ability.

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# Vision

To become Africa's leading AI-powered accessibility platform that transforms classrooms into inclusive learning environments where every student can learn, communicate, and participate equally.

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# Problem Statement

Millions of students experience barriers to learning because educational content is primarily delivered through speech.

Challenges include:

- Students with hearing impairments struggle to follow lectures.
- Institutions have limited access to qualified sign language interpreters.
- Existing captioning systems only transcribe speech without improving understanding.
- Most accessibility tools require continuous internet connectivity.
- There is limited support for Nigerian Sign Language and localized accessibility solutions.

These barriers reduce educational access, classroom participation, and learning outcomes.

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# Product Objectives

SignBridge AI aims to:

- Improve classroom accessibility.
- Support deaf and hard-of-hearing students.
- Enhance learning through AI-generated explanations.
- Enable communication between hearing and deaf individuals.
- Reduce dependence on human interpreters.
- Operate in low-connectivity environments.
- Promote inclusive education across Africa.

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# Target Users

### Primary Users

- Deaf students
- Hard-of-hearing students
- Sign language users

### Secondary Users

- Lecturers
- Teachers
- Schools

- Universities
- Inclusive education centres
- NGOs
- Healthcare workers

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# Core Features

| Feature | Description | Status |
|---|---|---|
| **Live Captions** | Converts lecturer speech into real-time captions for accessible learning. | ✅ Implemented (V0) |
| **AI Contextual Learning** | Generates explanations of concepts while the lecture is ongoing. | ✅ Implemented (V0) |
| **Visual Learning Cues** | Displays important concepts as interactive learning cards. | ✅ Implemented (V0) |
| **Smart Lecture Summary** | Produces concise AI-generated summaries for revision. | ✅ Implemented (V0) |
| **Saved Lessons** | Stores captions and summaries for future access. | ✅ Implemented (V0) |

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| **Accessibility Controls** | Adjustable text size and optimized reading experience. | ✅ Implemented (V0) |
| **AI Sign Language Recognition** | Detects hand gestures through the camera and translates them into text. | 🚧 In Development |
| **Sign-to-Speech Translation** | Converts recognized signs into spoken language. | 🚧 Planned |
| **Speech-to-Sign Translation** | Converts spoken words into animated sign language for deaf users. | 🚧 Planned |
| **Offline AI Processing** | Performs speech recognition and sign recognition without internet access. | 🚧 Planned |
| **Nigerian Sign Language Support** | Supports localized sign language datasets and recognition. | 🚧 Planned |
| **AI Tutor** | Answers questions and explains lecture concepts. | 🚧 Planned |
| **Automatic Quiz Generator** | Creates quizzes from lecture content. | 🚧 Planned |

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|---|---|---|
| **Lecturer Dashboard** | Enables lecturers to monitor accessibility and share resources. | 🚧 Planned |
| **Student Learning Analytics** | Tracks engagement, saved lessons, and learning progress. | 🚧 Planned |

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# User Journey

## Inclusive Classroom Experience

```
Lecturer Speaks
        │
        ▼
 Speech Recognition
        │
        ▼
  Live Captions
        │
        ▼
 AI Context Engine
    ┌─────────────────────┐
    │        │
Visual Cues   Smart Summary
    │        │
    └────────┬────────────┘
             ▼
 Accessible Learning
```

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## Sign Language Communication

```
Student Signs
      │
      ▼
Camera Input
```

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▼
Hand Detection
|
▼
Gesture Recognition
|
▼
Text Translation
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▼
Speech Output

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# Functional Requirements

The platform shall:

- Capture classroom audio in real time.
- Generate live captions.
- Identify and explain key concepts using AI.
- Generate lecture summaries.
- Save lecture notes and summaries.
- Allow accessibility customization.
- Detect sign language gestures.
- Translate signs into text.
- Convert recognized signs into speech.
- Convert speech into sign language animations.
- Operate both online and offline.
- Support multiple users.
- Integrate with educational environments.

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# Non-Functional Requirements

| Requirement | Target |
| --- | --- |

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|---|---|
| Caption Accuracy | >90% |
| Sign Recognition Accuracy | >90% |
| Response Time | <1 second |
| Offline Availability | Full support for core accessibility features |
| Security | Encrypted local and cloud storage |
| Scalability | Multi-classroom deployment |
| Accessibility Compliance | WCAG-aligned interface |

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# Technology Stack

### Frontend

- React
- TypeScript
- Tailwind CSS

### Artificial Intelligence

- OpenAI APIs
- Whisper (Speech-to-Text)
- MATLAB (Research & AI Model Development)
- Computer Vision Toolbox
- Deep Learning Toolbox
- TensorFlow Lite / ONNX Runtime (Offline Deployment)

### Backend

- FastAPI
- Node.js

### Database

- SQLite (Offline)
- PostgreSQL (Cloud)

### Mobile

- Flutter

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# Development Roadmap

### Phase 1

- Live captioning
- AI contextual learning
- Smart summaries
- Saved lessons
- Accessibility controls

### Phase 2

- Sign language recognition
- Sign-to-text translation
- Text-to-speech
- Offline AI support

### Phase 3

- Speech-to-sign translation
- Nigerian Sign Language support
- AI tutor
- Quiz generation
- Lecturer dashboard

### Phase 4

- Institutional deployment
- Learning analytics
- Learning Management System (LMS) integration
- Multi-language support

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# Success Metrics

- Caption accuracy above 90%.
- Sign language recognition accuracy above 90%.
- Reduced communication barriers in classrooms.
- Increased lecture comprehension and participation.
- Improved student satisfaction with accessibility tools.
- Adoption by educational institutions and inclusive learning centres.

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# Competitive Advantage

| **Existing Solutions** | **SignBridge AI** |
|---|---|
| Only speech captions | Live captions + AI learning support |
| One-way translation | Two-way communication support |
| Generic accessibility | Designed specifically for inclusive education |
| Internet dependent | Offline-first roadmap |
| No localized support | Nigerian Sign Language support |

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|---|---|
| Passive transcription | Context-aware AI explanations and summaries |
| No learning features | AI tutor, quizzes, lesson storage, analytics |

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# Long-Term Vision

SignBridge AI is envisioned as a comprehensive accessibility ecosystem for education. By combining real-time captioning, AI-assisted learning, sign language translation, offline intelligence, and localized support for African sign languages, the platform seeks to ensure that every learner can fully participate in the classroom regardless of hearing ability, communication preference, or internet availability. This unified approach aligns closely with the AFRETEC challenge's goal of using digital innovation to expand access, equity, and opportunity in education.